Rakibul Hasan Rafi

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Google Scholar | Researchgate | My Portfolio Website

EDUCATION

Bangladesh University of Engineering & Technology (BUET) | Dhaka, Bangladesh **[Jan 2022 to July 2026]**
B.Sc. in Chemical Engineering, CGPA: 3.39 / 4.00 (Last two terms average > 3.65 / 4.00)

Notre Dame College | Dhaka, Bangladesh **[2018 to 2020]**
Higher Secondary Certificate (HSC), Science | Result: GPA 5.00 / 5.00 (A+)

RESEARCH INTERESTS

- Sustainability & Environmental Development
- Wastewater Treatment
- Biomass Valorization, Biochar Synthesis & Toxicant Adsorption
- Bioprocess Engineering & Sustainable Chemical Processes
- Harmful Algal Blooms (HABs) & River Self Healing Spatial Modeling

TECHNICAL SKILLS & CERTIFICATIONS

Research, Writing & Office Skills

- **Technical Writing & Documentation:** Authored several reputated peer reviewed journal articles, 5 book chapters (submitted), 10 review papers, in-preparation research manuscripts, and open datasets; experienced in scientific report and technical proposal writing.
- **Research & Analytical Skills:** Literature review (Google Scholar), bibliometric analysis (VOSviewer), experimental design, statistical analysis, data visualization, and optimization techniques across sustainability and environmental engineering topics.

Data Analysis & Modeling Software

- Python (NumPy, Pandas, Matplotlib, Scikit learn), OriginPro (scientific graphing), Design Expert (RSM, CCD/BBD optimization)

Environmental & Process Engineering Software

- **Simulation Tools:** Aspen HYSYS, COMSOL Multiphysics, Visual MINTEQ (chemical equilibrium modeling), Design-expert

Machine Learning & AI

- Artificial Neural Networks (ANN, ANN GA, ANN PSO), Response Surface Methodology (RSM)

PUBLICATIONS & MANUSCRIPTS

Peer Reviewed Journal Publications

- Md. Mahmudur Rahman, Md. Tariqul Islam, Salah Knani, Farzana Mim, **Rakibul Hasan Rafi**, Orchi Datta, Reem Alreshidi, Pabittro Chandra Barman. *"Simultaneous elimination of both toxic heavy metals and hazardous dyes from wastewater bodies by CNC-induced ceramics membranes: A crucial overview on the advanced fabrication and applications."* **International Journal of Biological Macromolecules** (2025).
- Md. Mahmudur Rahman, Md Arafat Hossain Shourov, Salah Knani, Abdul Matin, Md Mostofa Habib, Reem Alreshidi, **Rakibul Hasan Rafi**. *"A critical review focusing on the production and application of GO-CNC-based nanosorbents in wastewater treatment: Removal of toxicants by fixed bed column adsorption study."* **Journal of Contaminant Hydrology** (2026).
- Md Ibrahim Shekh, M. S. I. Masum, **Rakibul Hasan Rafi**, Ishtiaq Murshed, Shadman Sakib Sayem, Azmain Akib Akash, Xiong Zhu, Sakil Mahmud. *"Adsorbent-modified membranes for selective contaminant removal - surface functionalization strategies and performance enhancement."* **Journal of Materials Science** (2026).
- Md. Abdur Rahim Miah, **Rakibul Hasan Rafi**, Tahzib Rayhan Himadry, Md Ibrahim Shekh, Md. Ashraful Alam, Raton Kumar Bishwas, Shirin Akter Jahan. *"Green Route Derived Crystallographic Phase and Biography Study on Nano-crystalline FCC Silver: A Perspective Crystallographic Review Insights."* **Cleaner Chemical Engineering** (2026).

- Tasnim Alam Subah, Hasan Tasnim Salehin, **Rakibul Hasan Rafi**, Md. Fozle Rabbi, Sayem Kawsar, Sumaiya Khan, Shahla Rahman, Sifat Kawsar, Md Zillur Rahman. *"Polymer Nanocomposites with Hybrid Nanoparticles: Architecture-Process-Structure-Property-Performance Perspectives."* **RSC Advances** (Accepted / In Press).

Open Datasets

- Md Ibrahim Shekh, **Rakibul Hasan Rafi**. *"Hourly Electricity Generation, Demand, Load Shedding, and Fuel Mix Dataset for Bangladesh (2015–2026)."* **Mendeley Data** (2026).

Submitted & Under Review Papers

- "Cathode materials from lignocellulosic biomass-derived CNC-AC-based biopolymeric nanosorbents after industrial wastewater purification: A state-of-the-art review" —Under Review
- "Functionalized Poly(vinylidene Fluoride) Membranes for Heavy Metal Removal: Mechanisms, Design Strategies, Sustainability, and Future Perspectives" —Under Review
- "A Review on Emerging Paradigms in Seawater Desalination: Integrating Advanced Membranes, Renewable Energy, and Smart Technologies" —Under Review
- "A comprehensive review of freeze desalination: Road to the future technology" — Submitted

Manuscripts in Preparation

- "Evaluating the Self-Healing Capacity of a Sewage-Impacted River: Spatial Analysis of Water Quality in the Buriganga River Near a Sewage Discharge Point"
- "Aminated chickpea-waste biochar for malachite green removal: Synthesis, characterization, mechanisms, and comparative optimization via RSM, ANN-GA, and ANN-PSO"
- "Toxic crystal violet removal by PA treated chickpea-waste valorized biochar: Synthesis, characterization, mechanisms, and comparative modelling by RSM, ANN, SVM"
- "Toxic HABs-bloom prediction with city of Defiance, Ohio"
- "HABs prediction from Maumee River watershed"

Book Chapters (Submitted)

- "Mitigating the Environmental Impact of Aerospace Materials: A Life Cycle Perspective" Chapter 22 in Multifunctional Composites and Smart Manufacturing for Aerospace Engineering (Ed. Yogesh Singla), Taylor & Francis
- "Computational Intelligence for Advanced Aerospace Materials" Chapter 13 in Multifunctional Composites and Smart Manufacturing for Aerospace Engineering (Ed. Yogesh Singla), Taylor & Francis
- "AI and ML Driven Biomaterials Design" Chapter 24 in Tumor Responsive Biomaterials: Smart Polymers, Hydrogels, and Advanced Drug Delivery for Cancer Therapy, Elsevier
- "Self-healing Natural Fiber Composites" — Chapter 11 in High Performance Natural Fiber Composites: Advancing Sustainable Materials for Modern Engineering (Ed. Sayantan Bhattacharya), Wiley
- "Introduction to Wood Science and Technology: Structure, Chemistry, Innovations in Wood Formulations, Processing and Engineering of Wood Products, Properties, Challenges, and Future Perspectives" Chapter 1, Elsevier

INDUSTRIAL EXPERIENCE

Shahjalal Fertilizer Company Limited (SFCL) | Sylhet, Bangladesh **[May 14 to 25, 2025]**

Industrial Trainee

- Conducted site visits to the industrial wastewater treatment plant, gaining practical insights into effluent management and treatment processes.
- Collaborated with multidisciplinary engineering teams to evaluate waste minimization and resource recovery strategies relevant to industrial scale process implementation.
- Gained hands on exposure to industrial safety protocols, environmental compliance, and large scale chemical/fertilizer manufacturing practices.

Samuda Chemical Complex Limited (SCCL) | Dhaka, Bangladesh

Industrial Visit Participant

- Gained practical exposure to large scale chemical manufacturing unit operations including hydrogen peroxide, caustic soda, liquid chlorine, hydrochloric acid, and sodium hypochlorite production.

TEACHING & ACADEMIC MENTORSHIP

Udvash Academic & Admission Care | Dhaka, Bangladesh **[2023 to 2025]**

Question & Answer Teacher (Teacher ID: 10259)

- Solved over 5,600 complex STEM problems for high school students across Physics, Chemistry, Mathematics, Biology, and Statistics.

Online & Private Tutoring | Dhaka, Bangladesh **[2021 to Present]**

Chemistry, Physics & Mathematics Tutor

- Tutored high school students in physical, organic, and inorganic chemistry, deep conceptual understanding and exam preparation.

CO CURRICULAR & VOLUNTEER ACTIVITIES

- **Rover Scout:** Active volunteer in university STEM outreach programs and community science communication.
- **Academic Competitions & Events:** International Astronomical Search Collaboration, International Day of Mathematics 2022, Climate Quiz 2023, Wall Magazine Exhibition 2019 (Notre Dame College), Chemistry Olympiad 2020.